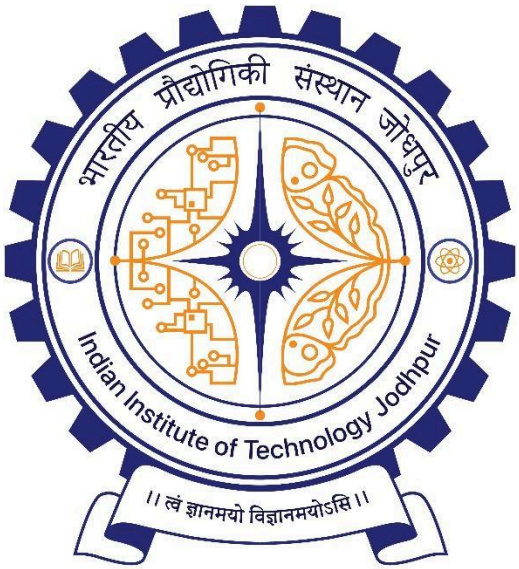
A new specialized PG degree program in the Department of Mechanical Engineering

M.Tech. in Engineering Design



Department of Mechanical Engineering
Indian Institute of Technology Jodhpur
2026-27
M.Tech. in Engineering Design

M. Tech. in Engineering Design (ED)

Department of Mechanical Engineering

Background

The rapid advancement of engineering systems in sectors such as mechanical design, aerospace, automotive, energy, defense, civil infrastructure, and advanced manufacturing has significantly increased the demand for engineers with deep expertise in mechanics and design. Modern engineering challenges involve complex material behavior, structural integrity, lightweight and high-performance designs, and reliability under extreme loading and environmental conditions. These challenges require advanced knowledge of continuum mechanics, computational modeling, finite element analysis, and design optimization, competencies that go beyond undergraduate-level training. Establishing an M.Tech. program in Engineering Design (ED) addresses this critical need by preparing graduates who can bridge the gap between fundamental mechanics and practical design engineering in high-technology industries and research organizations.

Furthermore, with the growing emphasis on indigenous technology development, Industry 4.0, additive manufacturing, and sustainable engineering, there is an urgent need for specialized postgraduate programs that integrate theory, computation, experimentation, and data-driven approaches. The proposed program at IIT Jodhpur is well-positioned to cater to national priorities by developing skilled manpower capable of contributing to advanced R&D, doctoral research, and innovation-led industries. The program will also strengthen interdisciplinary research and industry-academia collaboration, thereby supporting India's long-term goals in self-reliant manufacturing, infrastructure development, and cutting-edge engineering solutions.

Objectives of the Program

The objectives of the program are:

- To impart advanced knowledge of solid mechanics, elasticity, plasticity, viscoelasticity, fracture mechanics, structural dynamics, and continuum mechanics for analyzing and designing engineering systems.
- To develop competence in design engineering methodologies, including conceptual design, detailed design, optimization, and validation of mechanical and structural components and systems.
- To train students in modern computational techniques, including finite element analysis, numerical modeling, and simulation tools for solving real-world engineering problems.
- To enable students to integrate theoretical principles with experimental methods and computational modeling for robust engineering analysis and design.

- To inculcate strong research aptitude, critical thinking, and problem-solving skills, preparing students for doctoral studies, industrial R&D, and innovation-driven careers.
- To equip students with skills aligned with industry needs, including product design, structural integrity assessment, reliability analysis, and lifecycle engineering.

Graduate Attributes:

- Capability to identify, formulate, and analyze complex engineering problems using theoretical, computational, and experimental approaches.
- Proficiency in designing mechanical and structural systems considering functionality, manufacturability, safety, and sustainability through the application of advanced concepts of solid mechanics and design engineering.
- Ability to use modern computational tools such as FEA, optimization algorithms, simulation software, and machine learning for engineering analysis and design.

Learning Outcomes:

Students are expected to gain:

- A comprehensive understanding of fundamental and advanced principles of mechanics relevant to the analysis, design, and development of mechanical systems.
- Skills to design, analyze, optimize, and synthesize mechanical systems and components with consideration of performance, reliability, manufacturability, and real-world constraints.
- Specialized knowledge in advanced domains, including solid mechanics, vibrations, fatigue and fracture, biomechanics, control of mechanical systems, structural health monitoring, and the application of artificial intelligence in mechanical design.

Students are further expected to:

- Engage in hands-on laboratory work, simulations, and design projects to translate theoretical principles of solid mechanics and design engineering into realistic mechanical systems and structures.
- Develop industry- and research-ready competencies for careers in engineering firms, research and development organizations, government agencies, and academia, addressing challenges in mechanical system design, analysis, testing, and optimization.

Eligible Branches and Justification:

Regular students: B.Tech., B.E. (Mechanical Engineering, Production Engineering, Production and Industrial Engineering, Aerospace Engineering, Applied Mechanics, Aeronautical Engineering, Automobile Engineering, Ocean Engineering and Naval Architecture)

Sponsored students: B.Tech., B.E. (Mechanical Engineering, Production Engineering, Production and Industrial Engineering, Aerospace Engineering, Applied Mechanics, Aeronautical Engineering, Automobile Engineering, Ocean Engineering and Naval Architecture)

Student Intake Number and Strategy

A maximum of 15 students is planned to be admitted to this program. These include 8 GATE-qualified students (MoE fellowship), 7 sponsored students, and 2 sponsored students from research laboratories such as DRDO/ISRO.

Overall Program Philosophy

The M.Tech. program in Engineering Design at IIT Jodhpur is designed to develop strong foundations in mechanics, dynamics, computational methods, and design, with an emphasis on practical relevance and system-level thinking. The program prepares students to analyze, design, and optimize mechanical systems and structures to meet contemporary and emerging design engineering challenges.

This is enabled through (1) a core curriculum covering engineering mathematics, Theory of elasticity, dynamics and vibration of mechanical systems, finite element analysis, and advanced mechanical design to provide breadth and essential hands-on skills, and (2) a set of electives simultaneously offering depth and breadth together with application-oriented learning in advanced mechanics, dynamics, and data-driven design methodologies.

Objectives of the Core Program

The core program is designed to provide students with a strong foundation in fundamental and advanced principles of solid mechanics and design engineering, along with a deeper understanding of the analysis, modeling, and realization of mechanical systems and structures. This is achieved through a set of core theory courses, as listed below.

Objectives of the Elective Program

The elective structure of the proposed M.Tech. in Engineering Design is carefully designed to equip students with in-depth knowledge and practical skills across key areas critical to the field. It is to be noted that a total of 9 credits is to be completed through the program elective route. The list of program electives applicable M.Tech. in Engineering Design is provided below:

Course Structure:
Semester-wise Distribution of Credits

Cat.	*Course Number, Course Title*		*L-T-P*	*Credits*	*Cat.*	*Course Number, Course Title*		*L-T-P*	*Credits*
I Semester					**II Semester**				
C	MEL7XX0	Engineering Mathematics	3-0-0	3	C	*MEL7XX0*	Advanced Mechanical Design	3-0-0	3
C	MEL7XX0	Theory of Elasticity	3-0-0	3	E	MEL7XX0	Elective	3-0-0	3
C	MEL7XX0	Finite Element Analysis	3-0-0	3	E	MEL7XX0	Elective	3-0-0	3
C	MEL7XX0	AI, ML, and Big Data for mechanics and design	3-0-0	3	E	MEL7XX0	Elective	3-0-0	3
NG	LAN7010	Technical Communication	1-0-0	S/U	NG	OAL7010	*Innovation and IP Management*	1-0-0	S/U
			Total	**12**				*Total*	**12**
					Summer Term				
					T	MET7010	Thesis		8
								Total	8
III Semester					**IV Semester**				
T	MET7XX0	Thesis		16	T	MET7XX0	Thesis		16
			Total	**16**				*Total*	**16**

Program's Structure

S.No.	**Course Type**	**Credits**
1	Compulsory	15
2	Program Electives	9
3	M. Tech. Thesis I	16
4	M. Tech. Thesis II	16
5	M. Tech. Summer Thesis	8
Graded		**64**
6	Non-Graded Courses (2 credits)	0

Total		**64**

Elective Courses

Electives list

S. No.	Course Number	*Course Title*	*L-T-P*	*Credits*
1	MEL7XX0	Experimental Mechanics	3-0-0	3
2	MEL7XX0	Vehicle Engineering and Design	3-0-0	3
3	MEL7XX0	Engineering Fracture Mechanics	3-0-0	3
4	MEL7XX0	Acoustics	3-0-0	3
5	MEL7XX0	Dynamics and Vibrations of Mechanical Systems	3-0-0	3
6	MEL7XX0	Mechanics of Soft Materials	3-0-0	3
7	MEL7XX0	Nonlinear Finite Element Analysis	3-0-0	3
8	MEL7XX0	Sports Biomechanics	3-0-0	3
9	MEL7XX0	Rotor Dynamics	3-0-0	3

Title	**Theory of Elasticity**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.	Type	C
Prerequisite			

Objective:

1. To introduce the fundamental concepts of stress, strain, and constitutive relations in elastic solids and develop the mathematical framework required to analyze deformation and stress distribution in engineering materials and structures.
2. The course aims to provide analytical tools for solving boundary value problems in elasticity and to understand classical solutions involving torsion, bending, and energy methods.

Learning Outcomes:

1. Apply the concepts of stress, strain, and constitutive relations to analyze deformation and stress distribution in elastic solids. Determine principal stresses, shear stresses, and material response under different loading conditions.
2. Formulate and solve basic boundary value problems in linear elasticity and analyze torsion and bending of structural members. Apply energy principles and approximate methods to estimate stresses and deflections in engineering structures.

Contents:

Stress and Equilibrium [6 Lectures]: Cartesian tensors, Index notation, Internal forces and equilibrium, Traction vector and Cauchy's theorem, Stress tensor and its properties, Transformation of stress components, Normal and shear stresses, Principal stresses and principal directions, Maximum shear stress, Linear and angular momentum balance, Mohr's circle, Stress invariants, Decomposition of stress tensor, Octahedral stresses.

Kinematics of deformation and Strain [6 Lectures]: Local state of deformation at a material point, Displacement gradients and deformation tensor, Green-Lagrange Strain Tensor, Euler-Almansi Strain Tensor, Small strain tensor, Physical interpretation of strain components, Transformation of strain components, Strain invariants, Compatibility of strains.

Constitutive Equations and Material Failure [7 Lectures]: Linear elasticity and Hooke's Law, Elasticity tensor and transformation of basis, Symmetries of the stiffness tensor, Engineering or Voigt notation, Material symmetries and anisotropy, Isotropic linear elastic materials, Elastic constants (Young's modulus, Poisson's ratio, shear modulus, bulk modulus), Thermoelastic effects, Particular states of stress and strain of interest: Uniaxial stress, Uniaxial strain, Plane stress, Plane strain, Failure criteria: Maximum normal stress, Tresca Yield Criterion, Von Mises Yield Criterion, Octahedral shear stress, Distortion strain energy theory.

Boundary value problems in linear elasticity [7 Lectures]: Summary of field equations, Displacement formulation, Navier's Equations, Stress formulation, Beltrami-Michell Equations, Principle of superposition, Saint-Venant's Principle, Two-dimensional problems in elasticity, Plane stress problems, Plane strain problems, Airy Stress Function.

Torsion [5 Lectures]: Formulation of the basic equations of torsion of prismatic bars, Prandtl Stress Function, Torsion of noncircular bars, Warping function formulation, Membrane analogy, Torsion of bars with hollow thick-wall sections, Torsion of bars with thin hollow cross-sections.

Bending of Beams [4 Lectures]: Simple beam theory, Stress and strain distribution in pure bending, Generalization of simple beam theory to three dimensions and general cross sections, Combined effects of bending, shear, and torsion.

Energy Methods [4 Lectures]: Vector vs energy approaches to elasticity problems, Energy formulation of the elasticity problem, Minimum Potential Energy Principle, Ritz Method for approximate calculation of deflections and stresses in one-dimensional structural elements.

Text books

1. Timoshenko, S.P., and Goodier, J.N., 1980. Theory of Elasticity, Tata McGraw- Hill
2. Sadd, M. H. 2006. Elasticity: Theory, Applications and Numerics, Academic Press

Reference books

1. Sokolnikoff, I.S., 1956. Mathematical theory of elasticity, McGraw-Hill International.
2. Zhilun, X., 1992. Applied Elasticity, Willey Eastern Ltd.
3. Fung, Y.C., 1965. Foundations of Solid Mechanics, Prentice-Hall, Englewood Cliffs. New Jersey.
4. Srinath, L.S., 1980. Advanced mechanics of solids, McGraw-Hill International.

Self-learning

1. Theory of Elasticity, NPTEL, https://nptel.ac.in/courses/105105177
2. Advanced Strength of Materials, NPTEL, https://nptel.ac.in/courses/112101095

Title	**Finite Element Analysis**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.	Type	C
Prerequisite			

Objective: The objective is to

1. Develop a rigorous understanding of the finite element method (FEM) and its application to the numerical solution of differential equations arising in engineering problems.
2. Formulate, implement, and interpret results for scalar- and vector-valued problems, including time-dependent and eigenvalue analyses.

Learning Outcomes: Upon successful completion of this course, students will be able to:

1. Derive finite element formulations using direct and variational approaches.
2. Implement finite element formulations for bars, beams, heat transfer, and elasticity problems.

Contents

Introduction [4 Lectures]: Background; roadmap in FE-analysis, applications of finite elements; Direct approach for discrete systems: bar element, equation for assembly, boundary conditions, and system solution, applications to other linear systems, transformation laws, truss systems.

Variational approaches [5 Lectures]: Functional, variational operator, Euler-Lagrange equation, governing differential equations, and associated functional form. Rayleigh-Ritz Method, weighted residual approach, collocation method, least square method, Galerkin approach, weak formulation. Trial function, Test function.

Finite element formulation for 1D problems [4 Lectures]: FEM approach, Local approximation function and global approximation function, convergence, continuity, and completeness. Compatibility. Weak form for bar with varying cross-section, 1D linear shape function, Discrete equation, Local and global stiffness matrix, force vector from weak form. Numerical integration, Gauss quadrature, Transformation from physical space to parent space.

FE-formulation for Beam elements [4 Lectures]: Beam theory, governing differential equation and boundary conditions, Weak formulation, Hermite shape function, Derivation of local stiffness matrix, Force vector, numerical examples with beam and bar element.

FE-formulation for two-dimensional scalar-valued problems [4 Lectures]: Two-dimensional scalar-valued problem. Governing differential equation of the heat conduction with boundary conditions. Weak formulation, Approximation function, Pascal's triangle, Triangular element shape function derivation, FE-formulation of heat equation. Rectangular element shape function.

Isoparametric element and numerical integration [3 Lectures]: Isoparametric element. Shape function for a quadrilateral element. Transformation: physical space to parent space. Gauss quadrature in two dimensions. Compatibility and hanging nodes issue. Triangular element. Gauss quadrature points and weights for a triangular domain.

FE-formulation for two-dimensional vector-valued problems [4 Lectures]: Two-dimensional vector-valued problem. Governing differential equation elasticity equation. Plane stress and plane strain. Weak formulation, FE-formulation.

Finite element analysis of time-dependent problems [3 Lectures]: Transient heat conduction equation, Weak formulation, FE-formulation. Finite difference approximation for time. Explicit and implicit schemes.

Finite element analysis eigenvalue problems and structural dynamics [6 Lectures]: Eigenvalue problem, free vibration of a rod, free vibration of a beam. Structural dynamics, governing differential equations for a bar, including the inertia term. Weak formulation. FE-formulation. Consistent mass matrix and Lumped mass matrix. Explicit and Implicit time integration scheme. Newmark's method.

Textbooks:

1. Bathe, K.J., 2006. Finite element procedures. Klaus-Jurgen Bathe.
2. Cook, R.D., 2007. Concepts and applications of finite element analysis. John wiley & sons.

Reference books:

1. Fish, J. and Belytschko, T., 2007. A first course in finite elements (Vol. 1). New York: Wiley.
2. Reddy, J.N., 2026. An introduction to the finite element method. In *Dynamics of Earth's Fluid System* (pp. 199-226). CRC Press.
3. Hughes, T.J., 2003. The finite element method: linear static and dynamic finite element analysis. Courier Corporation.
4. Hutton, D., 2004. Fundamentals of finite element analysis. McGraw-Hill.
5. Seshu, P., 2003. Textbook of finite element analysis. PHI Learning Pvt. Ltd.

NPTEL material:

1. Introduction to Finite Element Method, NPTEL, https://nptel.ac.in/courses/112106135
2. Finite Element Method: Variational Methods to Computer programming, NPTEL, https://onlinecourses.nptel.ac.in/noc20_me60/preview

Title	**AI, ML and Big data for Mechanics and Design**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech./M.Tech.	Type	C
Prerequisite			

Objective: The objective is to develop an understanding of machine learning algorithms, and reduced-order modeling techniques for applications in solid mechanics and design.

Learning Outcomes:
1. Ability to implement machine learning and deep learning algorithms for prediction, analysis, and modeling in solid mechanics applications such as vibration, acoustics, fatigue, and fracture.
2. Ability to develop and apply AI-based computational frameworks using experimental and simulation data for design optimization, reduced-order modeling, and real-time monitoring.

Contents
Core Concepts in Machine Learning for Solid Mechanics (15 Lectures): Data Preprocessing, Feature Engineering, Linear Regression, Logistic Regression, Support Vector Machines, PCA, Dimensionality Reduction, Neural Network, Convolutional Neural Networks, Loss Functions, Evaluation Metrics, Naive-Bayes Classification algorithm, K-means clustering, K-nearest neighbours, Random forest, Bagging and Boosting

Reduced Order Models (4 Lectures): Proper Orthogonal Decomposition, Dynamic Mode Decomposition for analysis of vibration, acoustics, fatigue, and fracture.

Neural Network Architectures (10 Lectures): Physics Informed Neural Networks (PINNs), Custom Loss Functions with Physical Constraints, Recurrent Neural Networks (RNNs) and LSTMs, Autoencoders and GANs, Transformers, Applications in vibration analysis, acoustics, and fracture analysis

Experimental Data and ML Integration (4 Lectures): Sensor Data Interpolation for Real-Time Monitoring for vibration, acoustic, and fatigue analysis. Brief Introduction to Generative AI, Generative Methods for Design

Case Studies (6 lectures): Additional case studies based on the latest developments and not restricted to the above cases, Challenges, and future opportunities.

Text Books:

1. Bishop, C.M. and Nasrabadi, N.M., 2006. Pattern recognition and machine learning (Vol. 4, No. 4, p. 738). New York: Springer.
2. Kang, G., Kan, Q., Zhang, X., Hu, Y.N. and Li, X., 2026. Machine Learning in the Analysis of Deformation, Fatigue and Fracture in Solids. Elsevier.

Reference Books:

1. Bishop, C.M. and Bishop, H., 2025. Deep Learning.
2. Goodfellow, I., Bengio, Y., Courville, A. and Bengio, Y., 2016. Deep learning (Vol. 1, No. 2, pp. 1-800). Cambridge: MIT press.

3. Buyya, R., Calheiros, R.N. and Dastjerdi, A.V. eds., 2016. *Big data: principles and paradigms*. Morgan Kaufmann.

Title	**Advanced Mechanical Design**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.	Type	C
Prerequisite			

Objective:

1. To develop an understanding of mechanical design concepts under static, fatigue, fracture, reliability, and sustainability considerations.
2. To implement the developed concepts in real-life mechanical components and systems.

Learning Outcomes: Upon completion of this course, students will be able to:

1. Design mechanical components against fracture, fatigue, and creep failures under cyclic and high-temperature loading conditions.
2. Apply probabilistic methods for reliability-based mechanical design.
3. Incorporate sustainability principles and life-cycle thinking in engineering design.

Contents

Introduction to Design Methodology [3 Lecture]: Design philosophy, deterministic and reliability-based design, codes and standards in mechanical design, design variables, objective functions and constraints, safety factor, and design margins, Material selection in mechanical design; performance indices and Ashby plots.

Strength-Based Design under Static and Combined Loading [4 Lectures]: Failure theories for brittle materials: Maximum principal stress theory, Brittle Coulomb-Mohr Theory, Modified Mohr Theory; Failure for ductile materials: Maximum shear stress theory, Distortion energy theory, Ductile Coulomb-Mohr theory. Combined loading and failure envelopes.

Damage-Tolerant Design and Fracture Mechanics [9 Lectures]: Introduction to fracture mechanics, modes of fracture (Mode I, II, and III), Stress intensity factor, and Griffith criterion, linear elastic fracture mechanics, crack growth, and Paris law.

Fatigue and Creep Life Design [10 Lectures]: Introduction to fatigue failure, Different approaches to fatigue failure analysis, stress-life approach, strain-life approach, Fatigue under fluctuating stresses, Simple multiaxial fatigue, Cumulative damage, thermal fatigue; Creep behavior, Creep in high temperature low cycle fatigue.

Reliability-Based Design [4 Lectures]: Applications of statistics to material properties, probabilistic approach to design, and its comparison with safety factor approach, reliability prediction of series, parallel, and stand-by systems.

Design for Manufacturing [4 Lectures]: Selection of shapes, manufacturing processes. Design for assembly.

Sustainable and Life-Cycle Design [5 Lectures]: Introduction to Sustainability in Engineering, Sustainable Materials and Manufacturing, Circular Economy and Sustainable Design Strategies, Life Cycle Assessment (LCA) Methodology, Sustainability Metrics and Design Decision Making.

Text Books:

1. Hatamura, Y., 1999. The practice of machine design (Vol. 14). Clarendon press.
2. Norton R. L., 2010. Machine Design An Integrated Approach, Prentice Hall.

Reference Books:

1. Burr, A.H., 1995. Mechanical analysis and design, Elsevier.
2. Hatamura, Y., 1999. The practice of machine design (Vol. 14). Clarendon press.
3. Haugen, E.B., 1968. Probabilistic approaches to design, John Wiley.
4. Norton, R.L., 2002. Cam design and manufacturing handbook. Industrial Press Inc.

5. Kumar, P., 2009. Elements of fracture mechanics. McGraw-Hill Education LLC.
6. Boothroyd, G., Dewhurst, P. and Knight, W.A., 2010. Product design for manufacture and assembly. CRC press.
7. Agarwal AK, Biswas S. LCA and TEA for Sustainable Development. Springer Nature Singapore, Imprint: Springer; 2025 Oct 9.

Title	**Experimental Mechanics**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech./M.Tech.	Type	E
Prerequisite	Mechanics of Solids/Continuum Mechanics		

Objective: The objective is to:

1. Introduce the fundamental principles of the mechanics of solids and modern experimental techniques used to measure deformation, strain, and stress fields in materials and structures.
2. Apply fundamental concepts to measure real-life applications.

Learning Outcomes: Students will be able to

1. Understand the principles of measurement, instrumentation, and experimental methods used for stress and strain analysis in solids.
2. Apply measurement techniques to determine crack-tip displacement and residual stress.

Contents

Review of Mechanics of Solids [2 Lectures]: Concept of stress, strain, Equation of equilibrium, Stress-strain relation, Equation of compatibility.

Measurement principles and instrumentation [4 Lectures]: Measurement systems, static and dynamic measurement, calibration and sensitivity, error analysis and uncertainty quantification, data acquisition systems.

Contact Methods [6 Lectures]: Electrical resistance, Strain Gauges; Extensometers; Optical fiber strain gauges; nanoindentation.

Noncontact Methods [22 Lectures]: Image analysis and processing, Speckle methods, digital image correlation, Geometric Moire, Moire interferometry, Photoelasticity.

Applications [5 Lectures]: Fracture mechanics and crack-tip strain fields, Residual stress measurement, High strain rate and impact testing, Delamination mechanics.

Test Books:

1. Shukla, A. and Dally, J.W., 2014. Experimental solid mechanics (pp. 439-474). Knoxville, TN: College House Enterprises, LLC.
2. Ramesh, K. 2000. Digital Photoelasticity – Advanced Techniques and Applications. Springer-Verlag, Heidelberg, New York, Tokyo, 2000. ISBN: 3-540-66795-4.

Reference Books:

1. Sciammarella, C.A. and Sciammarella, F.M., 2012. Experimental mechanics of solids. John Wiley & Sons.
2. Sharpe Jr, W.N. and Sharpe, W.N. eds., 2008. Springer handbook of experimental solid mechanics. Springer Science & Business Media.
3. Ramesh, K., 2021. Developments in Photoelasticity: A renaissance. IOP Publishing.

Self-learning

1. Experimental stress analysis, NPTEL, https://onlinecourses.nptel.ac.in/noc21_me02/preview

Title	**Vehicle Engineering and Design**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech./M.Tech.	Type	E
Prerequisite			

Objective:

1. To cover the engineering design aspects of vehicles.
2. To appreciate the subsystem design issues such as Chassis, Steering, braking, and suspension, etc.
3. To understand the design trade-offs in vehicles.

Learning Outcomes:

1. To be able to design, simulate, and optimize vehicles.
2. To be able to apply industry standards for the design and evaluation of vehicles.

Contents

Introduction: History, vehicle classifications, fundamental approaches to vehicle modelling. [4 lectures]

Automotive Chassis design: Classification of chassis types, choosing of sections, Loads, Bending moment and Shear force diagrams, Design for Torsional rigidity, Light weight design strategy for vehicles. [4 lectures]

Vehicle Crashworthiness: Crash ratings of vehicles, Test facilities, Injury severity coding, Design of vehicle structures for crash energy absorption, Crash safety practices.[4 lectures]

Vehicle Occupant safety analysis: Design tools for human body, biomechanics of head, neck, thorax, Crash test dummy selection, Occupant protection measures. [4 lectures]

Acceleration Performance: Power train components; power-limited acceleration; traction limited acceleration; transverse weight shift; front wheel drive vs rear wheel drive vs. all-wheel drive vehicles. [4 lectures]

Braking Performance: Braking force analysis; brake design and analysis; federal regulation on braking performance; anti-lock braking system; wheel lock-up; tire/road friction; safety and maintenance issues in braking. [4 lectures]

Road Loads: Wind drag and car body design, rolling resistance; breakdowns of total road loads; gas mileage analysis and driving styles. [4 lectures]

Ride: Riding comfort; perception of vibration; vibration sources; vibration transmission to the passengers; vibration models; vibration isolation techniques, Tire specifications and constructions. [4 lectures]

Cornering/steering: Steering types and mechanisms, Lower speed cornering; high speed corner; cornering bicycle model; steering angle; over-steering vs. under-steering; cornering force analysis. [4 lectures]

Chassis control systems: Practical designs; active suspension, roll over and electronic stability program. [4 lectures]

Text Books:

1. An Introduction to Modern Vehicle Design Julian Happian-Smith, Butterworth Heinemann, 2002.

2. The Multibody systems Approach to Vehicle Dynamics, Mike Blundell and Damian Harty, Elsevier, 2004

Reference Books:

1. Road and Off-road vehicle system dynamics Handbook, Edited by G. Mastinu and M. Ploechl, CRC Press, Taylor & Francis Group, 2014.

2. Reimpell, Stoll and Betzler: The Automotive Chassis: Engineering Principles, Elsevier Science, 2001.

3. Fundamentals of Vehicle Dynamics, Thomas Gillespie, SAE Publication,

Multimdeia and online sources:

NPTEL Resources:

1. NPTEL :: Engineering Design - Vehicle Dynamics https://archive.nptel.ac.in/courses/107/106/107106080/

2. NPTEL :: Electrical Engineering - NOC:Fundamentals of Electric vehicles: Technology & Economics https://archive.nptel.ac.in/courses/108/106/108106170/

Title	**Engineering Fracture Mechanics**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech./M.Tech.	Type	E
Prerequisite	Mechanics of Solids		

Objective: The objective of the course is to

1. Provide the fundamentals of linear elastic fracture mechanics,
2. Introduce various numerical approaches for solving fracture problems.

Learning Outcomes: After completing this course, students will be able to

1. Solve fracture problems numerically using both discrete and diffuse approaches.
2. Interpret numerical results.

Contents

Background [3 Lectures]: Historical development, Review of crack-tip fields in linear elastic fracture mechanics (LEFM); Modes of fracture (I, II, III), Analytical solution of plane crack problems, stress intensity factors, energy release rate, and their equivalence.

Energy release rate [10 Lectures]: Griffith theory and extensions (Irwin, Orowan); Critical energy release rate; R-curve behavior and fracture stability, concept of J-integral (path independence), Relation between G and J, Introduction to elastic-plastic fracture mechanics (EPFM) parameters (CTOD, J as nonlinear energy release rate), calculation of SIF (analytical, experimental, numerical methods).

Review of Finite Element Method [5 Lectures]: Basic concepts of FEM, weak formulation of boundary value problems, discretization, interpolation functions, FE-formulation for scalar and vector valued problems, Assembly of global equations, and solution procedure.

Discrete approach for fracture simulation [13 Lectures]: FE for the cracked body, Need for mesh refinement near the crack tip, Singular and quarter-point elements, Method for determining fracture parameters, Domain and interaction integral approach. Extended finite element method, enrichment functions: crack tip and heavyside functions, Cohesive zone model.

Diffuse approach for fracture simulation [13 Lectures]: Introduction of various diffuse approaches, Phase-field model for fracture, derivation of governing equations from variational principle, Length-scale parameter and regularization; Phase-field for brittle fracture (AT model, phase-field evolution); Extensions to fatigue; Advantages over discrete methods (no explicit crack tracking, natural branching/coalescence); Numerical implementation (staggered/monolithic solvers).

Test Books:

1. Kumar, P., 2009. Elements of fracture mechanics. McGraw-Hill Education LLC.
2. Anderson, T.L. and Anderson, T.L., 2005. Fracture mechanics: fundamentals and applications. CRC press.

Reference Books:

1. Broek, D., 1982. Elementary engineering fracture mechanics. Springer Science & Business Media.
2. Kuna, M., 2013. Finite elements in fracture mechanics (Vol. 201). Dordrecht: Springer Netherlands.
3. Bordas, S.P., Menk, A. and Natarajan, S., 2023. Partition of unity methods. John Wiley & Sons

Self-learning:

1. Engineering Fracture Mechanics, NPTEL, https://nptel.ac.in/courses/112106065

Title	**Acoustics**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P-Th [C]	3–0–0 [3]
Offered for	B.Tech. (ME 3rd and 4th Year), M.Tech. and PhD	Type	E
Prerequisite	Linear Algebra and Calculus		

Objectives: This course aims to

1. Provide fundamental concepts of acoustics involving sound propagation, sound generation, sound radiation, sound reflection and transmission, and sound reception, and
2. Use of these concepts for studying and the development of noise control methods.

Learning Outcomes: The students are expected to have the ability to:

1. Understand characteristics of the acoustic field for different scenarios, like in a room, in a muffler, in a duct, in free field etc.
2. Identify different noise sources in a system and then, work to develop methods to reduce the noise.

Contents

Acoustic wave equation and its simple solutions [**3 Lectures**]

Spherical waves [**3 Lectures**]: Inhomogeneous acoustic wave equation, concept of point source

Reflection and transmission [**5 Lectures**]: Plane waves under normal and oblique incidence, Sound transmission through a fluid layer and a thin panel

Radiation [**5 Lectures**]: Sound field of oscillating spheres (as elemental sources), Sound field due to monopole, dipole and quadrupole sources, their importance, and their relation with oscillating spheres, sound field from a plane circular piston, continuous line source, radiation impedance, directivity and directivity index

Acoustic field in a rectangular waveguide of constant cross section [**3 Lectures**]

Pipes, resonators and filters [**4 Lectures**]: Resonance in pipes, power radiation from open ended pipes, Helmholtz resonator, Mufflers, Acoustic barriers and enclosures

Sound absorption [**5 Lectures**]: A simple model for growth of sound in a room, Reverberation Time, Sound absorption and Sound absorption materials, direct and reverberant sound

Electro-mechanical-acoustic analogies [**4 Lectures**]: Lumped parameter modeling of a loudspeaker using these analogies

Hearing [**2 Lectures**]: Anatomy of human ear and function of different parts, Fundamental properties of hearing

Noise and sound measurement [**2 Lectures**]: Introduction to sound measurement and instrumentation, frequency analysis, octave analysis, Combining band levels, Loudness level and loudness, Pitch and frequency

Green's function and Kirchoff's formula [**3 Lectures**]: Introduction to generalized functions and Green's theory for solution of partial differential equations, as required in acoustics, Reciprocal theorem in linear acoustics, Kirchhoff's formula of linear acoustics in the presence of surfaces.

Textbook

1. Kinsler L. E., Frey A. R., Coppens A. B., Sanders J. V., (1999), Fundamentals of Acoustics, 4th Edition, Wiley
2. Beranek L., Mellow T. (2019), Acoustics: Sound Fields, Transducers and Vibration, 2nd Edition, Academic Press

Reference books

1. Migeot J. L., Coyette J. P., Lielens G., (2016), Acoustics: Essential concepts, theory and models of linear acoustics for engineers, 1st Edition, IJK Numerics
2. Munjal M. L., (2013), Noise and Vibration Control, World Scientific Publishing Pte. Ltd.
3. Dowling and Ffowcs-Williams, (1983), Sound and sources of sound, Ellis Horwood Publishers.

Self-Learning Material

Acoustic and Noise Control, NPTEL, https://nptel.ac.in/courses/112106225/
Acoustics, NPTEL, https://nptel.ac.in/courses/112104026/
Acoustics, Teachingexcellence at MIT,
http://teachingexcellence.mit.edu/category/inspiring-teachers/amar-g-bose-6-312-acoustics

Title	**Dynamics and Vibration of Mechanical Systems**	Number	MEL7XX0
Department/ IDRP	Mechanical engineering	L-T-P [C]	3-0-0 [3]
Offered for	M. Tech.	Type	E
Prerequisite			

Objectives

1. To introduce the fundamental principles governing the dynamics of mechanical systems subjected to time-varying loads.
2. To develop the ability to formulate and analyze equations of motion for discrete and continuous mechanical systems using classical and variational methods.

Learning Outcomes: Students will be able to

1. Formulate equations of motion for mechanical systems using Newtonian and
2. energy-based approaches and analyze their dynamic behavior.

Contents:

Introduction [2 Lectures]: Overview and objectives of dynamics of mechanical systems. Types of dynamic loading. Essential characteristics of dynamic problems. Classification of mechanical systems: discrete and continuous systems. Distinction between rigid body dynamics and vibration analysis. Engineering applications of system dynamics.

Preliminaries of rigid body dynamics [6 Lectures]: Rigid body description and coordinate systems. Position and orientation representation. Rotation matrices and their properties. Homogeneous transformation matrices and coordinate transformations. Linear and angular velocity of a rigid body. Skew-symmetric angular velocity matrix. Space-fixed and body-fixed angular velocity descriptions. Linear and angular acceleration. Coriolis and centripetal acceleration. Mass distribution and inertia of rigid bodies. Properties of the inertia matrix and principal moments of inertia.

Formulations for system dynamics [6 Lectures]: Formulation of equations of motion for mechanical systems. Newton's laws and D'Alembert's principle. Generalized coordinates and generalized forces. Lagrange's equations of motion. Energy methods and functionals. Variational formulations in mechanics. Hamilton's principle and its application to mechanical system dynamics.

Dynamics of discrete systems [13 Lectures]: Single-degree-of-freedom (SDOF) system dynamics. Free vibration and forced vibration analysis. Response to harmonic, periodic, and impulsive excitations. Duhamel's convolution integral for dynamic response. Vibration isolation and transmissibility. Response to general dynamic loading. Fourier series representation of periodic loading. Multi-degree-of-freedom (MDOF) systems: formulation of equations of motion. Matrix representation of mass, damping, and stiffness. Modal analysis and modal superposition techniques.

Dynamics of continuous systems [12 Lectures]: Introduction to distributed parameter systems. Governing equations and boundary conditions. Free and forced vibration of strings. Longitudinal vibration of bars. Transverse vibration of beams. Torsional vibration of shafts. Analytical solution techniques for continuous systems. Approximate methods: Rayleigh's method and Rayleigh-Ritz method for estimating natural frequencies and mode shapes.

Textbooks:

1. Inman, D.J., 2017. Vibration with control. John Wiley & Sons.
2. Rao, S.S. and Yap, F.F., 1995. Mechanical vibrations (Vol. 4, pp. 75-848). New York: Addison-Wesley.

Reference Books

1. Meirovitch, L., 1975. Elements of vibration analysis, McGraw-Hill
2. Shabana, A.A., 2020. Dynamics of multibody systems. Cambridge University Press.
3. Kelly, S.G., 2011. Mechanical vibrations: theory and applications. Nelson Education.

Self learning:

1. Dynamics and Control of Mechanical Systems, NPTEL, https://onlinecourses.nptel.ac.in/noc22_me96/preview
2. Dymanics and Control I, MIT opencourseware, https://ocw.mit.edu/courses/2-003j-dynamics-and-control-i-spring-2007/

Title	**Mechanics of Soft Materials**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech. and B.Tech.	Type	E
Prerequisite			

Objective:

1. To introduce the nonlinear mechanics of soft materials such as rubbers, gels, and biological tissues, focusing on large deformations and nonlinear constitutive behavior.
2. The course develops the continuum mechanics and energy-based modeling framework needed to analyze instabilities, soft actuators, and failure in soft material systems.

Learning Outcomes:

1. Understand and apply nonlinear continuum mechanics and constitutive models to analyze large deformations and material behavior in soft materials.
2. Analyze mechanical instabilities, failure mechanisms, and soft actuators using energy-based and continuum modeling approaches.

Contents:

Introduction [7 Lectures]: Introduction to soft materials: rubber-like materials, polymer gels, elastomers, and soft biological tissues; characteristic features such as large deformations, nonlinear stress-strain behavior, and coupled physical responses. Mathematical preliminaries: physical quantities and their classification; scalars, vectors, and tensors; index notation; vector operations (dot, cross, and dyadic products); representation of partial differentiation using comma notation; gradient, divergence, and curl of tensors and their representation using index notation.

Nonlinear Continuum Mechanics [10 Lectures]: Kinematics of deformation: configurations and motion of continuum bodies, deformation mapping, deformation gradient, and strain measures for large deformations. Stress measures and balance laws: conservation of mass, balance of linear momentum, Cauchy stress tensor, stress measures for finite deformation, equations of motion, symmetry of stress tensor, balance of mechanical energy, and the principles of virtual work and power.

Hyperelastic Materials [8 Lectures]: Constitutive modeling of nonlinear elastic materials; objectivity; material symmetry and isotropic hyperelasticity; stress–deformation relations in terms of invariants; stress–deformation relations in terms of stretches; incompressible and compressible hyperelastic solids; different forms of strain energy density functions: neo-Hookean, Mooney-Rivlin, Ogden, Gent, Fung material, etc.; anisotropic hyperelastic materials and constitutive modeling.

Viscoelastic Materials [5 Lectures]: Coupling of elasticity and viscosity; kinematics of viscoelastic deformation; multiplicative decomposition of deformation gradient; rheological models; internal variable formulation; dissipation inequality and thermodynamic restrictions; constitutive and evolution equations.

Applications and Instabilities [9 Lectures]: Homogeneous deformation of hyperelastic membranes under uniaxial and biaxial loading; balloon inflation problem; energy method for the inflation of a spherical membrane; inflation of arteries under blood pressure; electroactive membranes; energy method for electroactive membranes; mechanical instabilities including snap-through instability and the Biot instability; morphological instabilities such as wrinkles, creases, folds, and ridges.

Text books

1. A. Freed, Soft solids, Birkhauser Verlag Ag, 2016.
2. K. Volokh, Mechanics of soft materials, Singapore, Springer, 2016.

Reference books

1. G.A. Holzapfel, Nonlinear Solid Mechanics - A continuum approach for engineering, 2000
2. L.R.G. Treloar, The physics of rubber elasticity, 3rd ed., 2005
3. P. Haupt, Continuum Mechanics and Theory of Materials, 2nd ed., 2002

Self-learning

Computational Continuum Mechanics, NPTEL https://nptel.ac.in/courses/112103296

Title	**Nonlinear Finite Element Method**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. /M.Tech.	Type	E
Prerequisite	Finite Element Method		

Objective: The objective is to

1. Introduce the fundamental concepts, mathematical formulation, and numerical implementation of the Finite Element Method (FEM) for solving engineering problems in solid mechanics, heat transfer, and related fields.
2. Develop practical skills in finite element modeling, computer implementation using MATLAB, and interpretation of numerical results for real-world engineering applications.

Learning Outcomes: On successful completion of this course

1. The student will understand the formulation and solution of geometrically and materially nonlinear structural problems, including stress-strain measures for large deformations, critical-point characterization, and incremental-iterative equilibrium procedures.

Contents:

Overview of Nonlinear Structural Analysis [5]: Introduction to nonlinear problems in structural analysis, including geometric and material nonlinearities, nonlinear forces, and nonlinear boundary conditions. Characteristics of force-deflection curves; equilibrium paths; identification of critical points such as limit points and bifurcation points.

Single Degree of Freedom Systems [6]: Single degree of freedom system exhibiting geometric nonlinearity. Formulation of nonlinear equilibrium equations. Incremental solution procedures. Iterative solution techniques using direct iteration and Newton-Raphson methods. Combined incremental iterative strategies, including full Newton-Raphson, modified Newton-Raphson, and initial stress (or initial stiffness) methods.

One-Dimensional Continuum Problems [5]: Axial bar under compression as a model problem. Kinematics of large deformation; various strain measures including engineering strain, Green-Lagrange strain, and Almansi strain. Weak and variational formulations based on the Green strain measure.

One-Dimensional Finite Element Formulation [7]: Total Lagrangian and Updated Lagrangian approaches. Derivation of nonlinear stiffness matrix and consistent tangent stiffness matrix. Analysis of limit points and bifurcation behavior; numerical strategies for traversal of critical points.

Two-Dimensional Nonlinear Problems [6]: Strain measures in two and three dimensions; deformation gradient tensor. Stress measures, including Cauchy stress and First and Second Piola-Kirchhoff stresses. Concept of objectivity and frame invariance. Updated Lagrangian formulation and stress increment procedures.

Two-Dimensional Incremental Finite Element Formulation [7]: Incremental formulation with configuration updates. Derivation of nonlinear stiffness matrix and consistent tangent stiffness matrix, including material and geometric stiffness contributions.

Advanced Solution Procedures [3]: Improvement of convergence using line search techniques. Arc-length methods for tracing equilibrium paths beyond limit points. Quasi-Newton and secant methods for enhanced computational efficiency.

Textbooks:

1. Belytschko, T., Liu, W.K., Moran, B. and Elkhodary, K., 2014. Nonlinear finite elements for continua and structures. John wiley & sons.
2. Kim, N.H., 2014. Introduction to nonlinear finite element analysis. Springer Science & Business Media.

Reference Books:

1. Kythe, P.K., Wei, D. and Okrouhlik, M., 2004. An introduction to linear and nonlinear finite element analysis: a computational approach. Appl. Mech. Rev., 57(5), pp.B25-B25.
2. Zienkiewicz, O.C. and Taylor, R.L., 2005. The finite element method for solid and structural mechanics. Elsevier.
3. Bonet, J. and Wood, R.D., 1997. Nonlinear continuum mechanics for finite element analysis. Cambridge university press.
4. Wriggers, P., 2008. Nonlinear finite element methods. Springer Science & Business Media.
5. Cook, R.D., 2007. Concepts and applications of finite element analysis. John wiley & sons.
6. Reddy, J.N., 2015. An Introduction to Nonlinear Finite Element Analysis: with applications to heat transfer, fluid mechanics, and solid mechanics. Oxford university press.

Title	**Rotor Dynamics**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.	Type	E
Prerequisite			

Objective:

1. To impart the fundamental concepts, principles, modeling, and analytical techniques of rotor dynamics, including transverse and torsional vibration behavior of rotating systems.
2. To develop an understanding of rotor-bearing system dynamics, condition monitoring, and performance assessment for the reliable operation of rotating machinery.

Learning Outcomes:

1. Demonstrate an understanding of the fundamental principles of rotor dynamics and apply analytical techniques to investigate the dynamic behavior of rotating machinery and rotor-bearing systems.
2. Apply rotor dynamic analysis, parameter identification, condition monitoring, and fault diagnosis techniques for performance evaluation, prognostics, and industrial problem-solving in rotating machinery.

Contents:

1. Review of dynamic principles using matrix algebra; Critical speeds and dynamic response; Stability of rotating components; Gyroscopic effects; Balancing of rotating machinery (**9 lectures**)
2. Torsional and transverse vibrations; Bearing dynamics and it's coefficient measurements (**8 lectures**)
3. Hydrodynamic Bearing Instabilities; Sub-Critical Phenomenon in Rotors (**8 lectures**)
4. Numerical and experimental studies employing a Jeffcott rotor; Fault Detection and analysis; Transducers and monitoring; Condition monitoring and signature analysis of Rotating Machines (**7 lectures**)
5. Development of complex rotor-dynamic models (**7 lectures**)

Text books

1. R Tiwari, Rotor Systems Analysis and Identification, CRC Press, Taylor & Francis Group, 2018.
2. Vance et al. Machinery Vibration and Rotor Dynamics, John Wiley & Sons

Reference books

1. Vance, J.M., 1991. Rotordynamics of turbomachinery. John Wiley & Sons.
2. Lee, C.W., 2012. Vibration analysis of rotors (Vol. 21). Springer Science & Business Media.

3. Goodwin M.J., Dynamics of Rotor-Bearing Systems, Unwin Hyman
4. Krämer E., Dynamics of Rotors and Foundations, Springer-Verlag

Self-learning

Theory & Practice of Rotor Dynamics, NPTEL https://nptel.ac.in/courses/112103024/

Title	**Sports Biomechanics**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.	Type	E
Prerequisite	Mechanics of Solids		

Objectives: The objective is to

1. Introduce the fundamental principles of biomechanics in sports and human movement.
2. To examine biomechanical factors contributing to sports injuries and their prevention.

Learning Outcomes: Upon successful completion of the course, students will be able to:

1. Explain the biomechanics of bones, joints, and muscles in sports movements.
2. Apply mechanical principles to analyse motion and forces in athletic activities.
3. Understand the biomechanical principles used in the design of sports equipment and protective gear.

Course Contents

Module 1: Introduction to Sports Biomechanics and Musculoskeletal System [8 Lectures]: Introduction, Importance of biomechanics in sports performance, Structure of the musculoskeletal system, Axial and appendicular skeleton, Types of joints and joint movements in sports, Muscle structure and function in human movement, Mechanical levers in the human body, Biomechanical analysis of major body segments (shoulder, elbow, spine, hip, knee, ankle).

Module 2: Mechanics of Human Motion in Sports[8 Lectures]: Statics in human movement, Dynamics in sports activities, Linear kinematics and angular kinematics of sports movements, Kinetics of human motion, Joint forces and muscle torque, Ground reaction forces in walking and running, Biomechanical analysis of running, jumping, and throwing.

Module 3: Mechanical Properties of Biological Tissues [8 Lectures]: Mechanical properties of bone, tendon, ligament, muscle and articular cartilage, Stress–strain behaviour of biological tissues, Hookean constitutive law, Linear mechanical behaviour, Nonlinear mechanical behaviour, Viscoelastic behaviour of biological tissues.

Module 4: Biomechanics of Tissue Loading and Injury [7 Lectures]: Mechanical stresses in connective tissues, Bone loading during sports activities, Tissue fatigue and

degeneration, Overuse injuries in sports, Impact loading and shock transmission, Anisotropic behaviour of biological tissues, Biomechanical principles of injury prevention.

Module 5: Biomechanics of Sports Equipment and Technology [8 Lectures]: Engineering design for sports performance, Athlete instrumentation and motion analysis, Wearable sensors in sports monitoring, Sports footwear biomechanics, Cushioning and energy return in sports shoes, Protective equipment and impact attenuation, Energy-absorbing materials in protective gear, Biomechanics of sports implements (racquets, bats, balls).

Textbooks:

1. Anthony J. Blazevich, Sports Biomechanics: The Basics – Optimizing Human Performance, Routledge.

2. Roger Bartlett, Introduction to Sports Biomechanics: Analysing Human Movement Patterns, Routledge.

Reference Books.

1. Fung, Y. C., Biomechanics: Mechanical Properties of Living Tissues, 2nd ed., Springer, New York, 1993.

2. Joseph Hamill, Kathleen M. Knutzen, Timothy R. Derrick. Biomechanical Basis of Human Movement. 4th Edition, Lippincott Williams & Wilkins, Philadelphia, USA, 2015.

3. James G. Hay, The Biomechanics of Sports Techniques, Prentice Hall.